

Spintronics Concepts and Symmetry

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1 Notation and Conventions

Unless a section explicitly states otherwise, vectors are bold, unit vectors carry a hat, and time is measured in seconds. Magnetic field strength $\mathbf{H}$ is in A m^{-1} , and effective flux density $\mathbf{B}_{\text{eff}}$ is in tesla, with $\mathbf{B}_{\text{eff}} = \mu_0 \mathbf{H}_{\text{eff}}$ when they describe the same effective interaction. The following symbols are used throughout the library.

$\mathbf{m}$ Normalized magnetization, with $\|\mathbf{m}\| = 1$.

$\mathbf{s}_i$ Unit spin direction at site i .

M_s Saturation magnetization of the ferromagnetic layer.

$\mathbf{H}_{\text{eff}}$ Effective magnetic field.

$\boldsymbol{\sigma}$ Spin-polarization unit vector.

α Dimensionless Gilbert damping parameter.

γ Positive gyromagnetic ratio, $\gamma = g\mu_B/\hbar$, where g is the Landé g -factor.

γ_0 Positive gyromagnetic factor for an $\mathbf{H}$ -field formulation, $\gamma_0 = \mu_0 |\gamma|$.

θ_{SH} Spin-Hall angle.

J_c Charge-current density.

t_{FM} Ferromagnetic-layer thickness.

β Signed torque ratio, $\beta = \tau_{\text{FLT}}/\tau_{\text{DLT}}$.

$\tau_{\text{DLT}}, \tau_{\text{FLT}}$ Damping-like and field-like torque amplitudes, respectively; both are in tesla in the LLGS equations.

e Positive elementary charge.

μ_0 Vacuum permeability.

μ_B Bohr magneton.

$\hbar$ Reduced Planck constant.

k_B Boltzmann constant.

T Absolute temperature.

Physical constants. The following values reproduce `constantfile.m`:

$$\begin{aligned}\gamma &= 1.760859644 \times 10^{11} \text{ rad s}^{-1} \text{ T}^{-1}, & e &= 1.602176565 \times 10^{-19} \text{ C}, \\ \mu_0 &= 1.25663706143592 \times 10^{-6} \text{ T m A}^{-1}, & \mu_B &= 9.274 \times 10^{-24} \text{ J T}^{-1}, \\ \hbar &= 6.58211951440 \times 10^{-16} \text{ eV s}, & k_B &= 1.38064852 \times 10^{-23} \text{ J K}^{-1}.\end{aligned}$$

Convention warning. The macrospin effective-field section of *Spin Dynamics Modeling Reference* defines both $\mathbf{H}_{\text{eff}}$ in A m^{-1} and $\mathbf{B}_{\text{eff}} = \mu_0 \mathbf{H}_{\text{eff}}$ in tesla. The atomistic Hamiltonian and LLGS sections use $\mathbf{B}_{\text{eff}}$ in tesla with γ in $\text{rad s}^{-1} \text{ T}^{-1}$, as do the macrospin and ASTRO implementations [33, 11]. Convert an $\mathbf{H}$ -field in A m^{-1} to its tesla-valued equivalent before using it in this LLGS equation; a code variable named `Hext` does not determine its units.

2 Spintronics Concepts

2.1 Antiferromagnets

An antiferromagnet is a magnetic material in which neighboring magnetic moments prefer an antiparallel arrangement. The opposing moments can compensate, giving little or no net magnetization even though the material has long-range magnetic order. Its order is therefore described more naturally by a Néel vector than by the net magnetization alone [9].

2.1.1 A-Type Antiferromagnet

An A-type antiferromagnet consists of ferromagnetic planes stacked antiferromagnetically: spins within each plane point in the same direction, all spins in the next plane point in the opposite direction, and the pattern repeats along the stacking direction. The word *plane* refers to the crystallographic planes selected by the exchange interactions in a particular material. Figure 1 chooses the z axis as the stacking direction as an illustrative convention, not as a universal crystal orientation.

Source status. Reference needed. This A-type structural classification was supplied directly for these notes and still needs a dedicated literature reference.

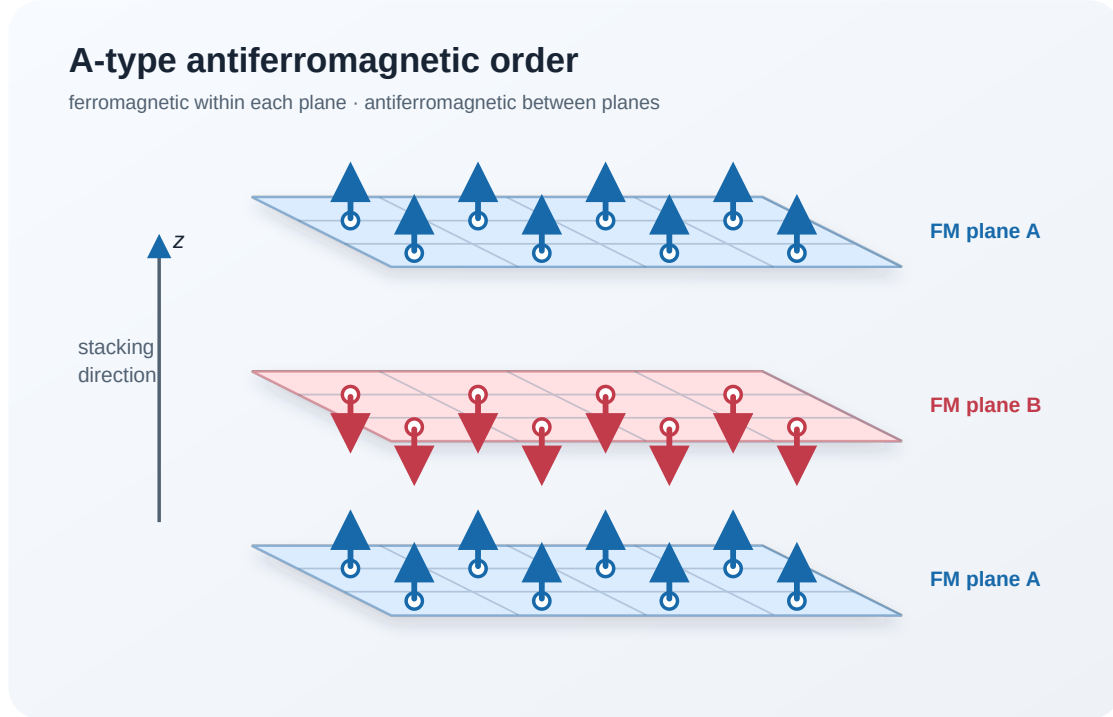


Figure 1: A-type antiferromagnetic stacking: ferromagnetic planes alternate in direction. Original schematic; the physical classification follows the provisional description above and therefore shares its **reference-needed** status.

The ferromagnetic bonds within each plane and antiferromagnetic bonds between planes are illustrated in Fig. 2. The exchange sign convention and pair counting for modeling this arrangement are given in the atomistic Hamiltonian of *Spin Dynamics Modeling Reference* [11].

For two equivalent alternating layers with normalized layer magnetizations $\mathbf{m}_A$ and $\mathbf{m}_B$, define the net and Néel order parameters as [9]

$$\mathbf{m} = \frac{\mathbf{m}_A + \mathbf{m}_B}{2}, \quad \mathbf{n} = \frac{\mathbf{m}_A - \mathbf{m}_B}{2}. \quad (2.1)$$

For an ideal compensated A-type state, these definitions give

$$\mathbf{m}_B = -\mathbf{m}_A, \quad \mathbf{m} = \mathbf{0}, \quad |\mathbf{n}| = 1. \quad (2.2)$$

The defining feature is therefore not simply zero magnetization. It is the specific spatial pattern of ferromagnetic order in two directions and antiferromagnetic alternation in the third. This distinguishes A-type order from G-type antiferromagnetism, in which every nearest neighbor is antiparallel, and C-type order, in which ferromagnetic chains are coupled antiferromagnetically. **Reference needed** for this final classification comparison.

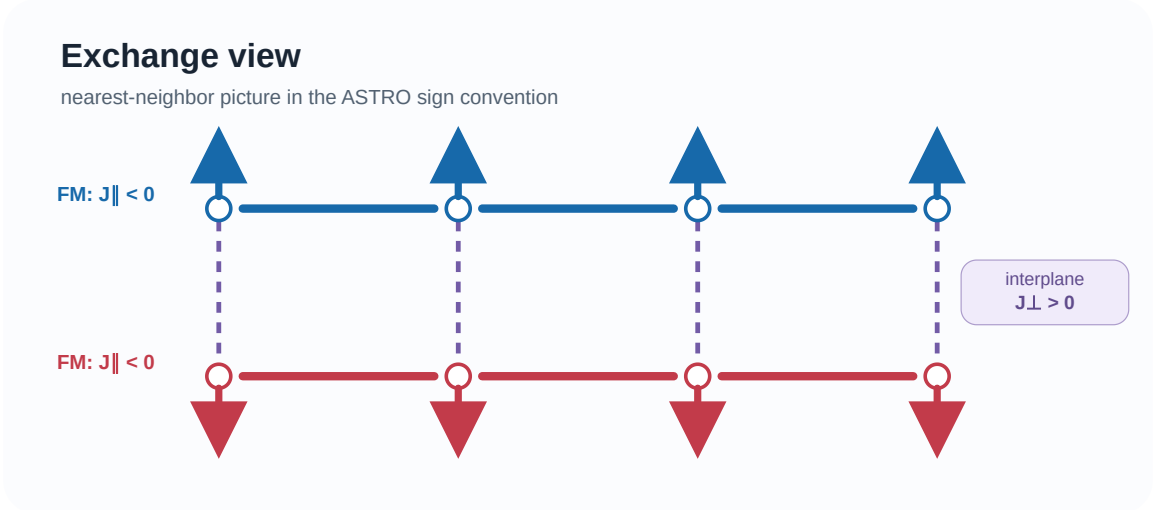


Figure 2: Exchange pattern for A-type order: parallel bonds within a layer and antiparallel bonds between layers. Original schematic based on the cited ASTRO exchange-sign convention [11]; the A-type classification itself remains **reference needed**.

3 Symmetry for Spintronics

A symmetry operation transforms the spatial coordinates, momenta, spins, and external fields of a system. It is a symmetry of a Hamiltonian only when the transformed Hamiltonian describes the same physical system. For a unitary operation G this condition is

$$G\mathcal{H}G^{-1} = \mathcal{H}, \quad (3.1)$$

whereas time reversal is antiunitary and also complex-conjugates scalar coefficients.

Source status. Reference needed. The general time-reversal and spatial-inversion operator rules in this section are included provisionally because no approved general reference is currently available. The mirror rules are supported by the EE237 notes, the isotropic-exchange example by ASTRO, and the magnetic parity–time example by Zhang *et al.*, as cited locally below.

3.1 Spatial Transformations and Rotation Matrices

This section uses column vectors and active transformations: the vector moves while the coordinate axes remain fixed. By the right-hand rule, a positive rotation about z is counterclockwise when viewed from the positive z axis. After a rotation through θ , the Cartesian basis vectors become

$$\begin{aligned} R_z(\theta)\hat{\mathbf{x}} &= \begin{pmatrix} \cos \theta \\ \sin \theta \\ 0 \end{pmatrix}, \\ R_z(\theta)\hat{\mathbf{y}} &= \begin{pmatrix} \cos(\frac{\pi}{2} + \theta) \\ \sin(\frac{\pi}{2} + \theta) \\ 0 \end{pmatrix} = \begin{pmatrix} -\sin \theta \\ \cos \theta \\ 0 \end{pmatrix}, \\ R_z(\theta)\hat{\mathbf{z}} &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \end{aligned} \quad (3.2)$$

To see why, note that a unit vector in the xy plane at azimuth ϕ has Cartesian components $(\cos \phi, \sin \phi, 0)^\top$. Initially, $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ have azimuths 0 and $\pi/2$, respectively. An active rotation

through θ adds θ to each azimuth, giving θ and $\pi/2 + \theta$; the identities $\cos(\pi/2 + \theta) = -\sin \theta$ and $\sin(\pi/2 + \theta) = \cos \theta$ then give Eq. (3.2). The vector $\hat{\mathbf{z}}$ is unchanged because it lies along the rotation axis [36]. Figure 3 shows the corresponding Cartesian projections.

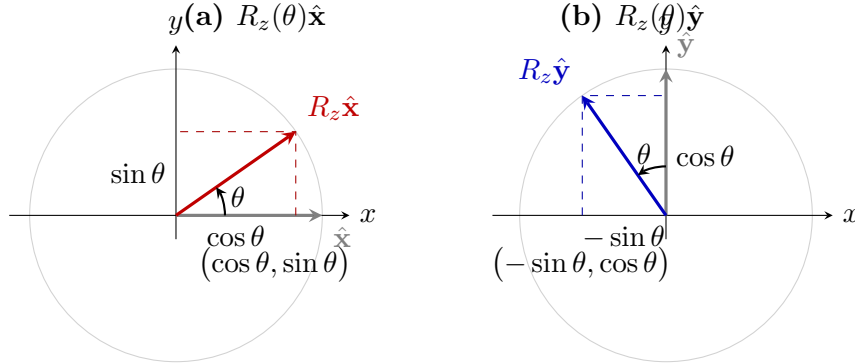


Figure 3: Original schematic showing how the Cartesian components of the actively rotated in-plane basis vectors are their projections onto the fixed axes. Both vectors rotate through the same angle θ and remain perpendicular; $\hat{\mathbf{z}}$, directed out of the page, is unchanged. The active-rotation convention follows the EE237 symmetry notes [36].

The columns of a transformation matrix are the transformed basis vectors. Therefore,

$$R_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad R_z(\theta) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \\ z \end{pmatrix}. \quad (3.3)$$

These conventions and the spatial transformation rules below follow the EE237 symmetry notes [36].

A passive rotation instead rotates the coordinate axes while leaving the physical vector fixed. It is represented by the inverse active rotation,

$$R_z^{-1}(\theta) = R_z^T(\theta) = R_z(-\theta). \quad (3.4)$$

Thus an unexplained sign difference between two rotation matrices often comes from using active and passive conventions, respectively [36]. Figure 4 summarizes the two viewpoints.

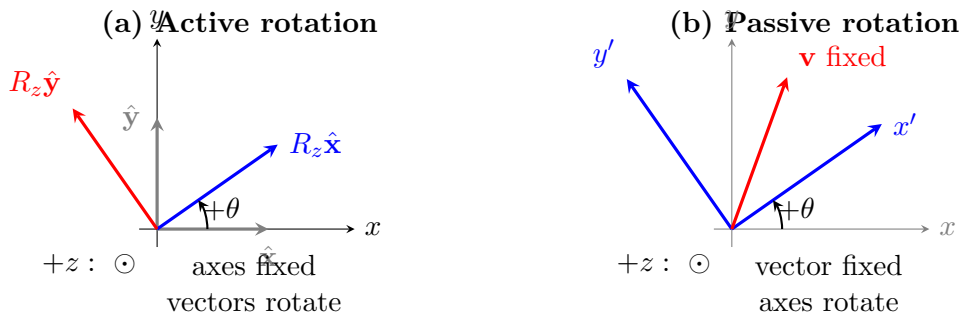


Figure 4: Original schematic comparing rotations about the z axis. (a) In an active rotation the coordinate axes remain fixed and the basis vectors rotate counterclockwise through $+\theta$; their images form the columns of $R_z(\theta)$. (b) In a passive rotation the physical vector remains fixed while the coordinate frame rotates through $+\theta$, so its coordinates transform with $R_z(-\theta) = R_z^T(\theta)$. The symbol $\odot$ denotes the positive z direction out of the page. The convention follows the EE237 symmetry notes [36].

A spatial transformation that preserves lengths and angles is represented by an orthogonal matrix,

$$R^\top R = I. \quad (3.5)$$

Taking determinants of both sides gives

$$\det(R^\top R) = (\det R)^2 = 1, \quad \det R = \pm 1. \quad (3.6)$$

The sign distinguishes whether the transformation preserves or reverses the handedness of the coordinate system. In three dimensions, a proper rotation preserves handedness and has $\det R = +1$. A mirror reverses one direction and has $\det R = -1$. Full spatial inversion is represented by $R = -I_3$ and also has $\det R = (-1)^3 = -1$ [36].

For example, setting $\theta = 90^\circ$ gives

$$\hat{\mathbf{x}} \rightarrow \hat{\mathbf{y}}, \quad \hat{\mathbf{y}} \rightarrow -\hat{\mathbf{x}}, \quad \hat{\mathbf{z}} \rightarrow \hat{\mathbf{z}}, \quad (3.7)$$

and hence

$$R_z(90^\circ) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad R_z(90^\circ) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -y \\ x \\ z \end{pmatrix}. \quad (3.8)$$

Expanding the determinant along the first row,

$$\det R_z(90^\circ) = 0 - (-1) \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + 0 = +1. \quad (3.9)$$

Thus the matrix is a proper rotation [36].

For comparison, reflection across the xy plane and full spatial inversion are represented by

$$R_{xy} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad R_{\text{inv}} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (3.10)$$

Since the determinant of a diagonal matrix is the product of its diagonal entries,

$$\det R_{xy} = (1)(1)(-1) = -1, \quad \det R_{\text{inv}} = (-1)(-1)(-1) = -1. \quad (3.11)$$

The inversion result depends on dimension: $\det(-I_n) = (-1)^n$. In two dimensions, inversion therefore has determinant $+1$ and is equivalent to a 180° rotation; in three dimensions it has determinant -1 [36].

3.2 Polar and Axial Vectors

Position $\mathbf{r}$, momentum $\mathbf{p}$, Bloch wavevector $\mathbf{k}$, electric field $\mathbf{E}$, and charge-current density $\mathbf{J}$ are polar vectors. Here $\mathbf{k}$ labels an electronic Bloch state in the Brillouin zone, $\hbar\mathbf{k}$ is the corresponding crystal momentum, and $\mathbf{k}$ is defined modulo a reciprocal-lattice vector. Orbital angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, spin $\mathbf{S}$, magnetization $\mathbf{M}$, and magnetic field $\mathbf{B}$ are axial vectors. For an orthogonal spatial transformation represented by R , the two vector classes transform as

$$\mathbf{v}' = R\mathbf{v}, \quad \mathbf{a}' = (\det R)R\mathbf{a}, \quad (3.12)$$

where $\mathbf{v}$ is polar and $\mathbf{a}$ is axial. Thus they transform in the same way under proper rotations ($\det R = +1$) and differently under mirrors and inversion ($\det R = -1$) [36].

The most frequently used transformations are summarized below. The entries describe the vector attached to the spatial point after that point has been mapped by the operation.

Operation	$\mathbf{r}, \mathbf{p}, \mathbf{k}$	$\mathbf{S}, \mathbf{M}$	$\mathbf{E}, \mathbf{B}$
Time reversal $\mathcal{T}$	$\mathbf{r} \rightarrow \mathbf{r}, \mathbf{p} \rightarrow -\mathbf{p}, \mathbf{k} \rightarrow -\mathbf{k}$	$\mathbf{S}, \mathbf{M} \rightarrow -\mathbf{S}, -\mathbf{M}$	$\mathbf{E} \rightarrow \mathbf{E}, \mathbf{B} \rightarrow -\mathbf{B}$
Inversion $\mathcal{P}$	$\mathbf{r} \rightarrow -\mathbf{r}, \mathbf{p} \rightarrow -\mathbf{p}, \mathbf{k} \rightarrow -\mathbf{k}$	$\mathbf{S}, \mathbf{M} \rightarrow \mathbf{S}, \mathbf{M}$	$\mathbf{E} \rightarrow -\mathbf{E}, \mathbf{B} \rightarrow \mathbf{B}$
Mirror $\mathcal{M}_{\mathbf{n}}$	$\mathbf{v} \rightarrow \mathbf{v} - 2\mathbf{n}(\mathbf{n} \cdot \mathbf{v})$	$\mathbf{a} \rightarrow -\mathbf{a} + 2\mathbf{n}(\mathbf{n} \cdot \mathbf{a})$	polar $\mathbf{E}$; axial $\mathbf{B}$

Source status. Reference needed. The Bloch-wavevector definition and its transformation rules require an approved project reference.

Figure 5 visualizes the different sign rules for polar and axial vectors under spatial inversion.

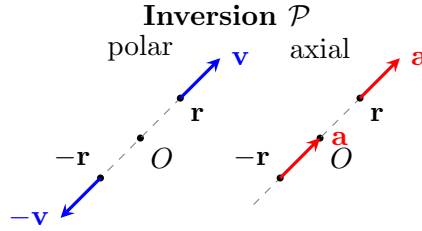


Figure 5: Original schematic of spatial inversion. Full inversion moves both vectors from $\mathbf{r}$ to $-\mathbf{r}$, reverses the polar vector, and preserves the axial-vector direction. The conventions follow the EE237 symmetry notes [36].

3.3 Time-Reversal Symmetry

Time reversal changes $t \rightarrow -t$ and reverses momenta and angular momenta. Spin is an intrinsic angular momentum, so

$$\mathcal{T}\mathbf{S}\mathcal{T}^{-1} = -\mathbf{S}, \quad \mathcal{T}i\mathcal{T}^{-1} = -i. \quad (3.13)$$

Here $\mathcal{T}$ is the time-reversal operator, not the superscript T used for a matrix transpose. Time reversal is antiunitary, so it complex-conjugates every scalar coefficient:

$$\mathcal{T}(c|\psi\rangle) = c^*\mathcal{T}|\psi\rangle. \quad (3.14)$$

Taking $c = i$, for which $i^* = -i$, and applying the operator product to an arbitrary state $|\phi\rangle$ gives

$$\mathcal{T}i\mathcal{T}^{-1}|\phi\rangle = -i\mathcal{T}\mathcal{T}^{-1}|\phi\rangle = -i|\phi\rangle. \quad (3.15)$$

Because this holds for every state, $\mathcal{T}i\mathcal{T}^{-1} = -i$. Thus the identity expresses complex conjugation by time reversal; it is not an ordinary matrix sign change. For example, in the position representation $\mathbf{p} = -i\hbar\nabla$, so complex conjugation reverses momentum:

$$\mathcal{T}\mathbf{p}\mathcal{T}^{-1} = i\hbar\nabla = -\mathbf{p}. \quad (3.16)$$

For a classical magnetization trajectory, the corresponding transformed trajectory is

$$\mathbf{m}_{\mathcal{T}}(t) = -\mathbf{m}(-t). \quad (3.17)$$

A magnetic state can therefore break time-reversal symmetry even when its nonmagnetic crystal structure does not. A fixed applied magnetic field, a fixed current, damping, or another nonequilibrium drive must not be silently left unchanged when testing time reversal; these time-odd or irreversible ingredients generally prevent the driven equation from being time-reversal-invariant.

Source status. Reference needed. Equations (3.13)–(3.17), including the antiunitary derivation and the general invariance statement above, still need an approved reference.

3.4 Parity or Spatial-Inversion Symmetry

Here parity means full spatial inversion,

$$\mathcal{P}: \quad \mathbf{r} \rightarrow -\mathbf{r}. \quad (3.18)$$

For inversion, $R_{\text{inv}} = -I_3$ and $\det R_{\text{inv}} = -1$. Substitution into the axial-vector rule in Eq. (3.12) gives

$$\mathbf{a}' = (\det R_{\text{inv}})R_{\text{inv}}\mathbf{a} = (-1)(-I_3)\mathbf{a} = \mathbf{a}. \quad (3.19)$$

Consequently, the shorthand $\mathbf{S}, \mathbf{M} \rightarrow \mathbf{S}, \mathbf{M}$ means that inversion does not reverse the components or direction of these axial vectors. It does move each vector from its original position to the inversion-related position:

$$(\mathbf{r}, \mathbf{S}(\mathbf{r}), \mathbf{M}(\mathbf{r})) \rightarrow (-\mathbf{r}, \mathbf{S}(\mathbf{r}), \mathbf{M}(\mathbf{r})). \quad (3.20)$$

Equivalently, when the transformed textures are written as functions of the coordinate $\mathbf{r}$,

$$\mathbf{S}_{\mathcal{P}}(\mathbf{r}) = \mathbf{S}(-\mathbf{r}), \quad \mathbf{M}_{\mathcal{P}}(\mathbf{r}) = \mathbf{M}(-\mathbf{r}). \quad (3.21)$$

This transformation law alone does not imply $\mathbf{S}(\mathbf{r}) = \mathbf{S}(-\mathbf{r})$ or $\mathbf{M}(\mathbf{r}) = \mathbf{M}(-\mathbf{r})$. Those equalities hold only when the complete magnetic texture is itself invariant under inversion. More generally, the structure has inversion symmetry only when the inverted sites, chemical species, interactions, and spin pattern reproduce the original structure. This is distinct from particle-exchange symmetry. In non-Hermitian coupled-mode models the symbol $\mathcal{P}$ may instead denote exchange of two subsystems; its meaning must therefore be stated explicitly.

Source status. Reference needed. Equations (3.18) and (3.19)–(3.21), including the axial-spin convention under inversion, still need an approved general reference.

3.5 Mirror Symmetry

A mirror with unit normal $\mathbf{n}$ reverses the perpendicular component of a polar vector. An axial vector behaves oppositely: its components parallel to the mirror plane reverse, while its perpendicular component is unchanged according to Eq. (3.12) [36]. For the xy mirror,

$$(v_x, v_y, v_z) \rightarrow (v_x, v_y, -v_z), \quad (S_x, S_y, S_z) \rightarrow (-S_x, -S_y, S_z). \quad (3.22)$$

The transformed lattice and every axial spin must both be compared with the original magnetic structure. A lattice may therefore possess a mirror plane that is broken by its magnetic order.

3.6 Spin-Rotation Symmetry

Spatial rotations move the lattice, whereas spin rotations act in spin space. For the isotropic exchange Hamiltonian used in ASTRO,

$$\mathcal{H}_{\text{ex}} = \frac{1}{2} \sum_{i \neq j} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (3.23)$$

a common spin rotation R_s leaves every dot product unchanged:

$$(R_s \mathbf{S}_i) \cdot (R_s \mathbf{S}_j) = \mathbf{S}_i \cdot \mathbf{S}_j. \quad (3.24)$$

The exchange term consequently has global spin-rotation symmetry when the couplings J_{ij} do not depend on spin direction [11]. Magnetic anisotropy, Dzyaloshinskii–Moriya interaction, dipolar coupling, an applied field, or another spin–orbit-coupled term can reduce this continuous symmetry to the subset of joint spatial and spin operations that preserves the complete Hamiltonian.

3.7 Combined Parity–Time Symmetry in a Two-Sublattice Magnet

For the two-sublattice order parameters defined in Eq. (2.1), suppose that inversion exchanges sublattices A and B . Since inversion does not reverse an axial spin, whereas time reversal does, their actions are

$$\begin{array}{c|cc} & \mathbf{m} & \mathbf{n} \\ \hline \mathcal{P} & \mathbf{m} & -\mathbf{n} \\ \mathcal{T} & -\mathbf{m} & -\mathbf{n} \\ \mathcal{PT} & -\mathbf{m} & \mathbf{n} \end{array} \quad (3.25)$$

For example, $\mathcal{P}$ maps $A \uparrow$ to $B \uparrow$, and $\mathcal{T}$ then maps it to $B \downarrow$, which is part of the original magnetic pattern. Thus a compensated antiferromagnetic pattern can be invariant under the combined operation even when it is not invariant under $\mathcal{P}$ or $\mathcal{T}$ separately. In a different use of parity, Zhang *et al.* define $\mathcal{P}$ as interchange of two coupled ferrimagnetic layers and obtain parity–time-symmetric Néel-vector dynamics at angular-momentum compensation [16]. These two meanings should not be conflated.

Source status. Reference needed. Equation (3.25) is a direct derivation from the provisional general transformation rules above. The cited paper supports only the separate coupled-layer example.

3.8 Mini-Tutorial: From $\mathcal{P}$, $\mathcal{T}$, and $\mathcal{PT}$ to Spintronic Effects

The useful question is not merely whether a crystal “has symmetry.” For a candidate response

$$O_i = \chi_{ij} F_j, \quad (3.26)$$

one must transform the observable O , the drive F , the magnetic order, and the complete device geometry. A symmetry of the driven structure is allowed to keep only tensor components that reproduce Eq. (3.26) after the transformation. Symmetry therefore decides which components may be nonzero, but not their magnitude or microscopic origin [21, 25, 26].

The signatures needed below follow directly from the polar/axial distinction:

Quantity	$\mathcal{P}$	$\mathcal{T}$	Character
$\mathbf{k}, \mathbf{E}, \mathbf{j}$	odd	$\mathbf{k}, \mathbf{j}$ odd; $\mathbf{E}$ even	polar
$\mathbf{S}, \boldsymbol{\sigma}, \mathbf{M}, \mathbf{B}$	even	odd	axial
spin current $Q_i^a \sim v_i S_a$	odd	even	polar flow index i , axial spin index a
Hall vector $\mathbf{h}$ in $\mathbf{j}_H = \mathbf{h} \times \mathbf{E}$	even	odd	axial response

The electric field is time-reversal even, whereas the dissipative current that it produces is time-reversal odd. A dc transport coefficient must therefore be interpreted as a nonequilibrium response, not as a new equilibrium order parameter.

Band-degeneracy checkpoint. From the transformation rules in Sec. 3.2, both $\mathcal{P}$ and $\mathcal{T}$ reverse $\mathbf{k}$, so their product acts within the same point of the Brillouin zone. For spin-1/2 electrons with $(\mathcal{PT})^2 = -1$, $\mathcal{PT}$ produces a Kramers-like partner at the same $\mathbf{k}$. Consequently, a $\mathcal{PT}$ -symmetric collinear magnet cannot have a same- $\mathbf{k}$ spin splitting. Breaking $\mathcal{PT}$ is necessary for such a splitting, but it is not sufficient: another unitary or antiunitary symmetry may still protect degeneracy [30, 32].

Source status. Reference needed. The response-tensor and $\mathcal{PT}$ band-degeneracy statements above are supported by the cited local reviews, but none of those reviews belongs to the project’s approved reference classes. They are retained provisionally rather than replaced by a weaker indirect source.

3.8.1 Rashba Spin–Orbit Coupling: Break $\mathcal{P}$, Keep $\mathcal{T}$

For an interface or polar crystal with polar axis $\mathbf{n}$, the leading isotropic momentum-linear term is

$$\mathcal{H}_R = \alpha_R(\mathbf{n} \times \mathbf{k}) \cdot \boldsymbol{\sigma}. \quad (3.27)$$

Time reversal changes both $\mathbf{k}$ and $\boldsymbol{\sigma}$ in sign, so Eq. (3.27) is $\mathcal{T}$ even.

Why the polar axis implies broken inversion. The axis $\mathbf{n}$ is a polar vector, such as an interface normal or the direction of a built-in electric field. Under inversion it transforms as $\mathbf{n} \rightarrow -\mathbf{n}$. If inversion were a symmetry of the same crystal, however, every fixed material property would have to be unchanged by that operation. A fixed polar axis would therefore have to satisfy $\mathbf{n} = -\mathbf{n}$, which permits only $\mathbf{n} = 0$. Equivalently, a nonzero $\mathbf{n}$ distinguishes the two opposite directions and inversion maps the crystal to a distinct structure or domain with axis $-\mathbf{n}$. Consequently, the Rashba term can be nonzero only when the environment breaks inversion symmetry, while time reversal may remain a symmetry. Typical examples are a surface, an asymmetric interface, or a polar crystal. The eigenvalues

$$\varepsilon_{\pm}(\mathbf{k}) = \varepsilon_0(\mathbf{k}) \pm \alpha_R |\mathbf{n} \times \mathbf{k}| \quad (3.28)$$

are split at a general $\mathbf{k}$ but obey $\varepsilon_+(\mathbf{k}) = \varepsilon_-(-\mathbf{k})$, as required by $\mathcal{T}$ [20].

Source status. Reference needed. Equations (3.27) and (3.28), together with the polar-axis inversion argument, still need an approved project reference.

3.8.2 Dzyaloshinskii–Moriya Interaction: Break Bond-Center Inversion

For one bond,

$$\mathcal{H}_{ij}^{\text{DMI}} = -\mathbf{D}_{ij} \cdot (\mathbf{S}_i \times \mathbf{S}_j). \quad (3.29)$$

The directed bond and DMI-vector convention are shown in the atomistic bond-interactions figure of *Spin Dynamics Modeling Reference* [11]. If inversion about the bond midpoint exchanges i and j , the axial spins do not reverse but their ordered cross product does:

$$\mathcal{P}: \quad \mathbf{S}_i \times \mathbf{S}_j \longrightarrow \mathbf{S}_j \times \mathbf{S}_i = -\mathbf{S}_i \times \mathbf{S}_j. \quad (3.30)$$

Invariance would then require $\mathbf{D}_{ij} = -\mathbf{D}_{ij}$, hence $\mathbf{D}_{ij} = 0$. DMI therefore needs the relevant local bond-center inversion to be absent; the whole crystal need not be globally polar. Under time reversal both spins change sign, leaving their cross product and DMI unchanged. DMI is consequently compatible with $\mathcal{T}$ and is selected primarily by spatial symmetry [11, 3].

3.8.3 Spin Hall and Anomalous Hall Effects: Similar Geometry, Different $\mathcal{T}$ Requirement

In an isotropic spin Hall material the leading response can be written

$$Q_i^a = \sigma_{\text{SH}} \epsilon_{iaj} E_j. \quad (3.31)$$

Here Q_i^a means a flow along direction i whose spins are polarized along direction a . A spin current therefore combines carrier motion, which is a polar vector, with spin, which is an axial vector. Under inversion the motion reverses while the spin does not, so Q_i^a reverses just as the electric field does. Under time reversal both the motion and the spin reverse, so their product Q_i^a is unchanged, just as the electric field is. Consequently, applying either $\mathcal{P}$ or $\mathcal{T}$ changes the two sides of Eq. (3.31) in the same way and does not require σ_{SH} to vanish.

An intuitive two-channel picture gives the same result: spin–orbit coupling deflects opposite spin polarizations toward opposite transverse directions. Their transverse charge currents cancel, but their spin currents add because both the spin and the direction of motion are reversed. Thus no net magnetization or broken inversion symmetry is required. Crystal mirrors and rotations can still force particular tensor components to vanish or relate them to one another, but they need not eliminate the spin Hall response as a whole. This is why a centrosymmetric nonmagnetic heavy metal can generate a transverse spin current [5, 6, 7, 22].

Worked example: magnetization-enabled spin-current components. Take an electric field E_x and ask which currents flowing along z are allowed. In the nonmagnetic geometry discussed by Davidson *et al.*, E_x preserves the mirrors σ_{xz} and σ_{xy} and the twofold rotation C_2^x . Testing the polar flow velocity and axial spin polarization under all three operations leaves only the conventional component Q_z^y allowed. A ferromagnetic order parameter lowers this symmetry: the operation remains a constraint only when it also preserves the chosen magnetization. The resulting selection rules are

Magnetization	Remaining sym- metry	Allowed responses shown	Newly allowed re- sponse
$\mathbf{m} \parallel \hat{\mathbf{x}}$	C_2^x	Q_z^y, Q_z^z	Q_z^z
$\mathbf{m} \parallel \hat{\mathbf{y}}$	σ_{xz}	j_z, Q_z^y	j_z
$\mathbf{m} \parallel \hat{\mathbf{z}}$	σ_{xy}	Q_z^x, Q_z^y	Q_z^x

For instance, when $\mathbf{m} \parallel \hat{\mathbf{x}}$, only C_2^x remains. This rotation reverses both the z -directed flow velocity and the z -directed spin, so their product Q_z^z is invariant. The two mirrors that forbade this component in the nonmagnetic system are no longer symmetries of the magnetized state. Thus magnetization can enable an anomalous-Hall charge current or a spin current with an unconventional polarization by removing the symmetry that previously forced that response to vanish. This test determines whether a component is allowed, not its sign, magnitude, or microscopic origin [22].

Source status for the worked example. Reference needed. The Davidson perspective supplies this example, but an approved project reference is still needed for the detailed ferromagnetic selection rules.

For the anomalous Hall effect,

$$\mathbf{j}_{\text{AH}} = \mathbf{h} \times \mathbf{E}, \quad (3.32)$$

where the Hall vector $\mathbf{h}$ is a compact representation of the antisymmetric Hall conductivity. Its transformation properties follow from the response equation. Both charge current and electric field are polar vectors, so inversion gives $\mathbf{j}_{\text{AH}} \rightarrow -\mathbf{j}_{\text{AH}}$ and $\mathbf{E} \rightarrow -\mathbf{E}$. The right-hand side then has the required minus sign only if $\mathbf{h}$ remains unchanged under inversion; this is the inversion rule for an axial vector. Time reversal instead gives $\mathbf{j}_{\text{AH}} \rightarrow -\mathbf{j}_{\text{AH}}$ because carrier velocity reverses, while $\mathbf{E} \rightarrow \mathbf{E}$. Therefore $\mathbf{h}$ must reverse under time reversal so that $\mathbf{h} \times \mathbf{E}$ also reverses.

In short, $\mathbf{h}$ is inversion even but time-reversal odd. Inversion does not forbid it, whereas an equilibrium state that preserves $\mathcal{T}$ must have $\mathbf{h} = 0$. Because $\mathcal{PT}$ also changes $\mathbf{h}$ in sign while leaving its spatial point fixed, $\mathcal{PT}$ forbids the anomalous Hall vector. Conversely, breaking $\mathcal{T}$ and $\mathcal{PT}$ is only a necessary screen: a mirror, rotation, or another antiunitary symmetry can still force a particular component of $\mathbf{h}$ to vanish. The Hall vector need not be proportional to a net magnetization, so compensated magnets can have an anomalous Hall effect when their magnetic point group allows a time-odd axial vector [29].

Source status. Reference needed. The anomalous-Hall symmetry criterion and Eq. (3.32) still need an approved project reference. Equation (3.31) is supported by the cited approved compact-model repository and the experimental references already used by this project.

3.8.4 Edelstein Effect: Uniform Versus Staggered Spin Accumulation

The Edelstein or inverse spin-galvanic response is

$$S_i = \chi_{ij} E_j. \quad (3.33)$$

Spin is inversion even and the electric field is inversion odd, so a global uniform tensor χ_{ij} must vanish when inversion is a symmetry. In a polar film with approximate $C_{\infty v}$ symmetry, the leading form is

$$\mathbf{S} = \chi_E \mathbf{n} \times \mathbf{E}. \quad (3.34)$$

The underlying material may preserve time reversal: the spin density is a nonequilibrium, current-odd response rather than an equilibrium magnetization.

Global inversion does not rule out a *staggered* Edelstein effect. If inversion exchanges two locally noncentrosymmetric magnetic sites, symmetry can impose

$$\mathbf{S}_A = -\mathbf{S}_B, \quad (3.35)$$

so that the unit-cell-average spin vanishes while a Néel-order-effective staggered field survives. This is the key distinction between global and local inversion asymmetry [25, 28, 21].

Source status. Reference needed. The uniform and staggered Edelstein equations need an approved project reference.

3.8.5 Spin–Orbit Torque: Convert a Symmetry-Allowed Spin Response into Torque

Once a current produces a spin polarization ζ , exchange coupling to a magnetic order permits the two familiar lowest-order torques

$$\tau_{\text{SOT}} = \tau_{\text{FL}} \mathbf{m} \times \zeta + \tau_{\text{DL}} \mathbf{m} \times (\mathbf{m} \times \zeta). \quad (3.36)$$

The device axes and the two torque vector factors are shown in the spin–orbit-torque geometry figure of *Spin Dynamics Modeling Reference* [5, 3]. The source symmetry determines ζ : a conventional isotropic bilayer gives $\zeta \parallel \mathbf{n} \times \mathbf{E}$, whereas lower crystal symmetry allows additional components. Field-like and damping-like angular forms do not uniquely identify Rashba/Edelstein and spin Hall microscopic mechanisms, because either mechanism can contribute to both forms [21, 4, 9].

A single vertical mirror illustrates how an unconventional component appears. Take the surviving mirror to be the yz plane, represented by $R = \text{diag}(-1, 1, 1)$. Because $\mathbf{E}$ is polar, it transforms as $\mathbf{E}' = R\mathbf{E}$, whereas the axial spin polarization transforms as $\zeta' = (\det R)R\zeta$. Hence $E_x \rightarrow -E_x$, $E_y \rightarrow E_y$, and $\zeta_z \rightarrow -\zeta_z$.

The selection rule follows by writing the possible linear response as $\zeta_z = \chi_{zx}E_x + \chi_{zy}E_y$. Mirror symmetry requires the transformed quantities to obey the same response law. The term $\chi_{zx}E_x$ satisfies this requirement because both E_x and ζ_z change sign, so $\zeta_z \propto E_x$ is allowed. In contrast, E_y is unchanged while ζ_z changes sign. The relation $\zeta_z = \chi_{zy}E_y$ can then equal its mirror image only if $\chi_{zy} = 0$, so an in-plane current parallel to the mirror plane cannot generate this component. Since ζ_z is normal to the bilayer plane, inserting this symmetry-allowed polarization into Eq. (3.36) produces the corresponding out-of-plane torque component. This is the symmetry logic behind the current-direction-dependent out-of-plane torque in WTe₂/ferromagnet bilayers [36, 27, 26].

For antiferromagnets, global inversion can forbid the uniform torque while local inversion asymmetry permits opposite sublattice torques. Whether the even or odd response is staggered depends on the operation that exchanges the sublattices; local noncentrosymmetry is necessary but not sufficient, so the full site and magnetic point groups must be checked [25, 28].

3.8.6 Spin-Transfer Torque: No Intrinsic Inversion-Breaking Requirement

Spin-transfer torque transfers angular momentum from a spin-polarized current whose polarization $\mathbf{p}$ is prepared by a magnetic polarizer, magnetic texture, or spin filter:

$$\boldsymbol{\tau}_{\text{STT}} = u_{\text{DL}} \mathbf{m} \times (\mathbf{p} \times \mathbf{m}) + u_{\text{FL}} \mathbf{m} \times \mathbf{p}. \quad (3.37)$$

Unlike Rashba SOC, DMI, or a uniform Edelstein response, conventional STT does not require spin-orbit coupling or broken crystal inversion. Its essential ingredients are a nonequilibrium spin flux and a noncollinear receiving magnetization. Magnetic order or spin filtering supplies the time-reversal-odd polarization $\mathbf{p}$, while reversing the current reverses the angular-momentum flux. Device geometry can of course break inversion, but that is not the microscopic selection rule for STT [5, 13, 12].

3.8.7 Altermagnetic Spin Splitting: Break $\mathcal{T}$ Without $\mathcal{PT}$ Degeneracy

In a conventional collinear antiferromagnet, opposite-spin sublattices are often related by inversion or a translation. A $\mathcal{PT}$ symmetry then protects same- $\mathbf{k}$ degeneracy directly. In the nonrelativistic spin group, a spin flip combined with a sublattice translation can likewise make the two spin channels degenerate throughout the Brillouin zone; after spin-orbit coupling is included, translation-time-reversal alone generally protects degeneracy only at selected momenta. An altermagnet instead relates the opposite-spin sublattices by a proper or improper crystal rotation $\mathcal{R}$, but not by a translation or inversion based spin-group operation that would restore general same- $\mathbf{k}$ degeneracy. Schematically,

$$\varepsilon_{\uparrow}(\mathbf{k}) = \varepsilon_{\downarrow}(\mathcal{R}\mathbf{k}), \quad \varepsilon_{\uparrow}(\mathbf{k}) \neq \varepsilon_{\downarrow}(\mathbf{k}) \quad \text{at a general } \mathbf{k}. \quad (3.38)$$

The rotation redistributes the two spin channels so that the net moment can still compensate. The resulting even-parity d -, g -, or i -wave spin splitting is nonrelativistic and does not require spin-orbit coupling; spin-orbit coupling subsequently modifies nodal degeneracies and enables additional Berry-curvature responses. Inversion may still exist as an operation within a same-spin sector, but it must not exchange the two opposite-spin sublattices in the way that restores $\mathcal{PT}$ degeneracy [30, 31, 32].

Source status. Reference needed. Equation (3.38) and the altermagnetic symmetry classification still need an approved project reference.

3.8.8 Selection-Rule Summary

Effect	Spatial-symmetry gate	Time-reversal / combined-symmetry gate
Rashba SOC	Global inversion broken; a polar axis is allowed	$\mathcal{T}$ may be preserved; $\mathcal{PT}$ degeneracy is absent
DMI	Inversion absent at the relevant bond midpoint	$\mathcal{T}$ may be preserved
Spin Hall effect	$\mathcal{P}$ may be preserved; point group selects tensor components	$\mathcal{T}$ may be preserved
Anomalous Hall effect	$\mathcal{P}$ may be preserved; remaining mirrors and rotations select $\mathbf{h}$	$\mathcal{T}$ broken and no $\mathcal{PT}$ or other symmetry that reverses the allowed Hall vector
Edelstein effect	Uniform response needs global inversion breaking; staggered response needs local inversion breaking	Equilibrium $\mathcal{T}$ breaking is not required for the driven spin accumulation
SOT	Spin-source and device point group must allow ζ ; local asymmetry can give staggered AFM SOT	Magnetic target breaks $\mathcal{T}$; torque parity alone does not identify the source mechanism
STT	No intrinsic $\mathcal{P}$ -breaking or SOC requirement	Requires a nonequilibrium spin flux; magnetic polarizer/filter supplies time-odd $\mathbf{p}$
Altermagnetic splitting	Opposite-spin sublattices rotation-related, not inversion- or translation-related	$\mathcal{T}$ broken; no $\mathcal{PT}$ or translation/inversion based spin-group symmetry that enforces general same- $\mathbf{k}$ degeneracy

This table gives necessary symmetry gates, not sufficient material-design conditions. After passing the gate, band filling, spin-orbit strength, scattering, interfaces, magnetic domains, and device geometry determine whether the effect is measurable or useful.

3.9 Practical Symmetry Test

To test a proposed symmetry of a spin model:

1. map every position and sublattice label;
2. transform spin as an axial vector and reverse it additionally under time reversal;
3. transform external fields, currents, and boundary conditions;
4. rebuild every Hamiltonian or dynamical term with the transformed quantities; and
5. compare the complete transformed model and magnetic pattern with the originals.

Checking only the atomic lattice or only the spin arrows is insufficient for a magnetic symmetry.

Source status. Reference needed. The general five-step test is a provisional pedagogical synthesis and still needs an approved reference.

4 Generating Unconventional Spin-Orbit Torque

Take the film normal as $\hat{\mathbf{z}}$ and drive an in-plane electric field $E_x \hat{\mathbf{x}}$. Write Q_z^a for a spin current flowing toward the receiving magnetic layer along z , with spin polarization along a . The conventional spin Hall geometry produces Q_z^y ; a nonzero Q_z^z instead supplies an out-of-plane spin

polarization, often called a σ_z component. The spin-current component, its transmission across the interface, and the torque measured in the receiver are distinct quantities. The damping-like torque from a transmitted σ_z component has the angular form $\mathbf{m} \times (\hat{\mathbf{z}} \times \mathbf{m})$, up to the sign of its amplitude [36, 5, 4]. This torque vanishes at exact collinearity, $\mathbf{m} = \pm \hat{\mathbf{z}}$; the subsequent switching dynamics must therefore be evaluated for the actual initial state and other torques [4]. The examples below describe ways to *allow or generate* this component; symmetry alone does not set its magnitude or establish that switching occurs.

4.1 Lower the Crystal Symmetry

A low-symmetry spin source can permit an out-of-plane torque for one in-plane current direction and forbid it for another. For example, with a surviving yz mirror, E_x and the axial spin component s_z both reverse, whereas E_y does not. A z -directed flow is unchanged by that mirror, so $Q_z^z \propto E_x$ is allowed while $Q_z^z \propto E_y$ is forbidden. WTe₂/ferromagnet bilayers exhibit this current-direction-dependent out-of-plane torque; the crystal orientation therefore matters when specifying the device axes [36, 27].

4.2 Use Magnetic Order to Rotate the Spin Polarization

The magnetization of a spin-source ferromagnet can remove symmetries that forbid spin-current components in a nonmagnetic source. Davidson *et al.* consider E_x and the operations σ_{xz} , σ_{xy} , and C_2^x . With no magnetization, only Q_z^y among the three z -flowing spin components survives all three operations. For a source magnetization $\mathbf{m}_s \parallel \hat{\mathbf{x}}$, only C_2^x remains: it reverses both the z flow and the z spin, making Q_z^z symmetry-allowed. The spin-rotated term in the perspective can be written schematically as

$$\mathbf{Q}_z^R \propto \mathbf{m}_s \times (\hat{\mathbf{z}} \times \mathbf{E}), \quad \mathbf{m}_s \parallel \hat{\mathbf{x}}, \quad \mathbf{E} \parallel \hat{\mathbf{x}} \implies Q_z^z \neq 0 \text{ is allowed.} \quad (4.1)$$

Reversing $\mathbf{m}_s$ reverses this term. This is a statement about the source's allowed spin-current polarization, not a prediction of its size or of the receiver's switching outcome [22].

Source status. Reference needed. The detailed ferromagnetic selection rule and Eq. (4.1) come from the Davidson perspective and its processed explanation; an approved project reference is still needed for these details.

A CoFeB/Ti/CoFeB trilayer provides a device example: the bottom in-plane magnetized layer or its interface generates both in-plane and out-of-plane spin-polarization components, and the upper perpendicular CoFeB layer is the receiver. Reversing the bottom-layer magnetization reverses the reported switching polarity [23].

Source status. Reference needed. The trilayer result has no approved project reference yet.

4.3 Design Electrical Asymmetry in a Polycrystalline Source

Broken electrical symmetry can also produce a z -polarized spin current without relying on a low-symmetry single crystal. Liu *et al.* report z -spin damping-like and field-like torques in sputtered polycrystalline heavy-metal devices with transverse and perpendicular electrical asymmetry. They vary the device length, width, thickness, and metal to tune the response, and demonstrate nearly full field-free switching of perpendicular FeCoB. The device geometry is therefore part of this generation route [24].

Source status. Reference needed. The electrical-asymmetry mechanism and device result have no approved project reference yet.

In each route, report the in-plane current direction, film normal, source magnetization where present, and receiving layer before assigning a sign to σ_z or to its torque amplitude [36, 5].

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